

KRISHNA GOJE

Staff Data Platform Engineer | Data Platform Architecture, Leadership & AI-Augmented Engineering

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PROFESSIONAL SUMMARY

Staff Data Platform Engineer with 15 years building enterprise data platforms at American Express, Grubhub, Booking.com, and Opendoor — including 8+ years managing engineering teams of 8 to 30+, with technical-track titles spanning Senior Manager and Director-band equivalents. Deep expertise in Apache Spark (architecture, performance optimization, distributed computing), AWS-native data stacks, and large-scale ETL/streaming pipeline design. Pioneered an agentic AI engineering ecosystem connecting 15+ enterprise platforms with parallel agent orchestration, RAG pipelines, and self-learning feedback loops — compressing week-long workflows into under one hour. At Opendoor, architected the operational data layer powering 70%+ of company dashboards, consolidated 100+ datasets into 3 unified models (\$300K/year saved), and built the data pipeline behind an AI VoiceBot projected to save \$5M annually. Track record of accelerating customer and business outcomes through scalable data infrastructure — turning pipeline reliability, real-time data access, and AI automation into measurable velocity gains for product, operations, and finance teams. Equally effective as a technical lead architect or people manager — consistent track record of building high-impact data platforms and managing engineering teams from 8 to 30+ people.

IMPACT BEYOND ROLE

- **Awarded 2 formal recognition awards** at Opendoor within first year from Sr. Director and Director of Operations for platform impact and delivery velocity
- **Credited by name in company-wide announcement** for architecting the data pipeline behind Opendoor's AI VoiceBot launch, projected to save \$5M annually
- **Granted autonomous authority** by Opendoor's Head of Data to deprecate and redefine company-wide operational metrics, a responsibility typically reserved for senior leadership
- **Solved Spark's small file problem** on American Express's Open-Source platform, publishing optimization patterns adopted by engineering teams across the 60,000-person enterprise
- **Organized technical and cultural events** engaging 1,000+ colleagues at American Express; published internal technical blogs on Spark and Hive optimization
- **Recognized as Hackathon Finalist (2017) and Data Geek award winner (2018)** for innovation in data engineering approaches at American Express

AI-AUGMENTED ENGINEERING PLATFORM | BUILT AT OPENDOOR | 2024–PRESENT

Pioneered a first-of-its-kind agentic AI engineering ecosystem that connects 15+ enterprise platforms into a unified, self-improving system — turning one engineer into a high-velocity data team. Built production RAG pipelines, multi-agent orchestration frameworks, and self-learning feedback systems that reduced week-long analytics workflows to under one hour while simultaneously improving data quality, consistency, and stakeholder response times.

- **Ecosystem Architecture:** Designed and connected 15+ Model Context Protocol (MCP) integrations — Snowflake, Databricks, QuickSight, dbt, Airflow, Slack, Glean, GitHub, Linear, Select Star, Google Workspace, Gmail, Google Calendar, and a custom semantic memory server — into a single AI orchestration layer. Built custom workarounds and API bridges where native MCP connectors did not exist, establishing end-to-end handshake mechanisms across the entire enterprise data stack
- **Parallel Agent Orchestration:** Engineered a multi-agent execution framework where 2–3 AI sub-agents work simultaneously, each running up to 8 parallel data warehouse queries. A 24-query cross-market

- investigation completes in under 60 seconds. Built 76 domain-specific automation skills that auto-detect task type and load specialized execution strategies without manual invocation
- **Self-Learning Feedback System:** Designed 9 automation hooks that capture error patterns, winning approaches, and domain knowledge at session end, then automatically update system configuration. The platform compounds in speed and accuracy with every use — mistakes are never repeated, optimizations are permanent, and institutional knowledge accumulates infinitely via a custom semantic memory layer
 - **Production Automation Library → Open Source ([quicksight-mcp](#)):** Built a 4,874-line Python library (146 functions) for programmatic BI dashboard management — then extracted and open-sourced as an MCP server with 31 tools, 59 tests, and a self-learning engine. Eliminated manual AWS console operations across 6+ production executive dashboards
 - **Autonomous Stakeholder Support → Open Source ([slack-data-bot](#)):** Deployed an L1 data support bot that monitors team communication channels, auto-investigates data questions using parallel warehouse queries, drafts responses through a Writer/Reviewer quality loop, and delivers polished answers. Open-sourced with configurable data sources and human-in-the-loop safety

EXPERIENCE

Staff Data Platform Engineer (Sr. Manager-band equivalent) | Opendoor (NYSE: OPEN) | Aug 2024 – Present

Data Platform & Executive Analytics

- Architected end-to-end operations data platform spanning home assessment through resale, serving as the single source of truth for 70%+ of operational decision-making across the company
- Consolidated 100+ fragmented datasets into 3 unified analytical models on Snowflake Data Warehouse and Databricks Lakehouse (Delta Lake) following Medallion Architecture, cutting BI compute and storage costs by 60% (~\$300K/year in measured savings)
- Built 6 executive dashboards powering Weekly Business Reviews, Sales Performance, and Operations Planning used by VP and C-level stakeholders daily
- Overhauled 8+ core dbt data models, resolving 800+ data quality issues in assessment classification that had silently corrupted downstream reporting
- Implemented pipeline observability layer across all milestone tables using dbt — automated freshness checks, schema validation, statistical tests, and dependency checks ensuring data integrity at every pipeline stage before stakeholder consumption
- Eliminated 7+ manual spreadsheet dependencies from production dashboards, removing human-error risk from executive reporting workflows
- Partnered with 15+ cross-functional stakeholders across Operations, Finance, Sales, Customer Success, and Engineering to define metric definitions, reporting standards, and data-driven strategies for customer retention and revenue growth
- Earned autonomous authority from Opendoor's Head of Data to deprecate and redefine company-wide operational metrics, reflecting deep trust in platform judgment and domain expertise
- Reporting directly to the Sr. Director of Data, own analytics engineering strategy with a Data Platform as a Product mindset — setting modeling standards, driving cross-team platform adoption, and developing engineers across dbt, Snowflake, and BI

AI Products & Innovation

- Architected data pipeline for Ernie AI VoiceBot, targeting 80% of seller leads that human agents never reach — projected \$5M annual savings from automated lead qualification; personally credited by name in company-wide launch announcement
- Built AI-generated executive summary system using Snowflake Cortex, delivering COO-level operational narratives from raw metrics without manual analyst intervention

Lead Data Engineer (Manager-band equivalent) | Booking.com (via Tavant Technologies) | Jan 2023 – Aug 2024

- Architected end-to-end data ingestion framework migrating 15+ financial reporting pipelines from legacy MySQL to AWS cloud-native architecture using DMS, Glue, Lambda, S3 (with cross-region replication for disaster recovery and lifecycle tiering for cost optimization), Step Functions, CloudWatch, DynamoDB, and Athena (Presto-compatible) — reducing report generation latency by 40% and improving data freshness from daily to near-real-time
- Built automated data observability and reliability framework with centralized log aggregation in CloudWatch, automated alerting and anomaly detection, multi-stage data quality checks (freshness, completeness, accuracy) across 50+ financial KPIs — established data SLOs and freshness SLAs, detecting ingestion anomalies and reconciliation discrepancies before reaching downstream audit surfaces
- Partnered with Finance and Accounting teams to translate multi-currency reconciliation logic into scalable SQL models on Snowflake, replacing fragile Excel-based workflows and ensuring regulatory compliance for financial reporting pipelines
- Implemented incremental loading patterns and partition pruning strategies that reduced pipeline compute costs by 35% while maintaining sub-hour data freshness
- Managed a cross-functional team of Java developers and data engineers; built and maintained a Java Spring Boot-based RESTful API service deployed on Kubernetes, serving as the orchestration layer between user-facing applications and AWS data infrastructure — routing event-driven requests to trigger DMS replication tasks, Glue jobs, and Lambda functions, enabling dynamic pipeline execution at scale

Lead Data Engineer | Grubhub | Nov 2021 – Dec 2022

- Managed a cross-functional engineering team building the Universal Listing Directory, aggregating and normalizing restaurant data from Google, Uber Eats, DoorDash, and internal sources into a single canonical dataset serving 300K+ restaurant partners
- Built Kafka-based event-driven ingestion pipelines processing multi-source restaurant metadata with real-time deduplication and entity resolution, powering operations dashboards used by 200+ internal stakeholders
- Provided sales teams with unified competitive intelligence on listing coverage, enabling data-driven restaurant onboarding and incentive targeting at scale

Lead Data Engineer (Manager-band equivalent) | American Express | Jan 2017 – Nov 2021

- Managed and mentored a team of 8–10 engineers, owning architectural decisions, technical hiring, and end-to-end delivery accountability across fraud detection, customer analytics, and marketing attribution programs
- Designed and built the feature engineering pipeline for Next-Gen Membership Rewards fraud detection, computing 2,000+ behavioral features from 40+ data sources, contributing to a system that prevented an estimated \$100M+ in annual fraudulent charges
- Tripled Apache Spark (PySpark/Scala) ETL throughput on distributed computing clusters through advanced data skew mitigation (salting, bucketing, adaptive partitioning), partition optimization, shuffle tuning, and memory management — reducing critical batch windows from 6+ hours to under 2; deep expertise in Spark internals, DAG optimization, and cluster resource management
- Solved Spark's small file problem and published memory optimization patterns on American Express's internal Open-Source platform, addressing industry-wide performance bottlenecks adopted by engineering teams across the 60,000-person enterprise
- Orchestrated 40+ production data pipelines using Apache Airflow with PagerDuty-integrated incident alerting, managing task dependencies, SLA monitoring, on-call incident response, and failure recovery across fraud detection, customer analytics, and marketing attribution workstreams
- Enforced enterprise data governance standards for PII data handling across 120M+ card member records — implementing multi-tier classification of sensitive attributes (SSN, driver's license, passport, name, address, mobile number) in compliance with regulatory and data privacy requirements

- Designed real-time event streaming pipeline processing attribution signals across 50+ marketing channels — connecting ad impression, click, and conversion events to customer profiles at scale across 120M+ card members
- Built Customer Engagement analytics layer measuring cardholder lifecycle metrics (activation, spend velocity, attrition risk) across 120M+ card members
- Published technical blogs on Spark and Hive optimization for internal knowledge sharing, establishing reusable best practices referenced by data engineering teams enterprise-wide
- Organized technical and cultural events engaging 1,000+ colleagues across the organization, building cross-functional community beyond core engineering responsibilities
- Earned Hackathon Finalist recognition (2017) and Data Geek award (2018) for innovation in data engineering approaches

Data Engineer | American Express (via Scepter Technologies) | Apr 2015 – Jan 2017

- Built Real-Time Federated Query Engine enabling analysts to query across NoSQL databases (HBase, Cassandra, Elasticsearch) through a unified SQL interface, reducing cross-platform data access time from days to minutes
- Developed query optimization and caching layers that improved federated query performance by 10x for repeated analytical workloads

Graduate Research Assistant — Data Engineering | East Texas A&M University | Aug 2013 – Dec 2014

- Conducted data engineering-focused research and coursework — implemented data processing pipelines using Hadoop, MapReduce, and SQL; built database systems projects deepening expertise in distributed computing, query optimization, and data modeling — establishing the technical foundation that directly informed subsequent industry data engineering roles

Software Engineering Intern (Java) | EnhanceEdu Research Lab, IIIT Hyderabad | Jan 2012 – May 2012

- Developed a Java-based project management tool deployed to students at IIIT Hyderabad's EnhanceEdu Research Lab — built end-to-end during undergraduate final-semester internship, applying object-oriented design, database integration, and user-facing application development on a real research-lab deployment

OPEN SOURCE

- [quicksight-mcp](#) — The most comprehensive AWS QuickSight MCP server. 31 tools, 59 tests, self-learning engine. Born from managing 6+ executive dashboards in production. Python, MIT License.
- [slack-data-bot](#) — Autonomous data Q&A bot with 8-strategy Slack monitoring, parallel AI investigation agents, and Writer/Reviewer quality loop. Human-in-the-loop safety. Python, MIT License.

TECHNICAL SKILLS

Data Platforms: Snowflake, Databricks (Delta Lake, Unity Catalog), ClickHouse, Apache Iceberg, AWS (EC2, S3, EMR, Redshift, Athena/Presto, DMS, Glue, Lambda, RDS, DynamoDB, Kinesis, Lake Formation, Step Functions, CloudWatch), Azure, GCP, Apache Spark (PySpark, Spark Streaming), Hadoop (HDFS, MapReduce, Hive, Oozie, Zookeeper, Storm), Trino, Kafka, SQL Server, PostgreSQL, Oracle, MySQL

Data Engineering: dbt, Apache Airflow, SQL (advanced), Python, Java, Scala, Spring Boot, Gradle, Shell Scripting, ETL/ELT, CDC (Debezium, AWS DMS), Streaming Pipelines, Data Lakehouse Architecture, Data Governance, Data Mesh, Domain-Driven Design, Agile/Scrum, JIRA, Rally

BI & Analytics: Amazon QuickSight, Tableau, Power BI, Apache Superset, Snowflake Cortex (LLM), Data Modeling, Dimensional Modeling, Data Warehousing

AI/ML & Automation: Agentic AI, LLM Integration, RAG Pipelines, Vector Search, Semantic Retrieval, Vector Databases (Chroma, FAISS), Prompt Engineering, AI Agent Development, LangChain, LangGraph, Hugging Face, OpenAI APIs, MCP (Model Context Protocol), Pydantic

Infrastructure: Terraform, Kubernetes, Docker, CI/CD, Git, GitHub Actions, Grafana, Splunk, PagerDuty, ServiceNow

EDUCATION

Master of Science, Computer Science | East Texas A&M University | GPA: 3.52 | 2014

Bachelor of Technology, Information Technology | JNTU Hyderabad | 2012

CERTIFICATIONS

- Databricks Certified Associate Developer for Apache Spark
- AWS Certified Solutions Architect – Associate
- Oracle Certified Professional, Java SE 6 Programmer